

FORMALIZATION OF LANGLANDS’S FIRST MAIN LEMMA FOR LOCAL EPSILON FACTORS

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ABSTRACT. We formalize in Lean 4 Langlands’s First Main Lemma for local epsilon factors over all nonarchimedean local fields. This equality of local constants in cyclic extensions of prime degree is one of the four Main Lemmas used in Langlands’s construction of epsilon factors of local Weil representations. The proof follows the author’s companion paper, which reconstructs and completes the local arguments of Dwork and Langlands, including the equal-characteristic case.

The First Main Lemma also follows from Deligne’s construction of local epsilon factors by global methods; here we formalize a purely local proof. The development includes the finite-sum and Haar-integral constructions of local constants, the required ramification and finite-field identities, and the comparison of stationary representatives under twisting and norm.

ChatGPT and Codex assisted with parts of the Lean development. The author used ChatGPT to check the manuscript for consistency and to suggest improvements to wording and organization of materials.

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1. INTRODUCTION

1.1. The First Main Lemma. Let E be a nonarchimedean local field with Weil group W_E . Langlands's local epsilon factors attached to representations of W_E are determined by their values on one-dimensional representations, multiplicativity, and compatibility with induction. Using Brauer induction, their construction is reduced to local epsilon factors of quasi-characters. One must then prove that the answer does not depend on the chosen Brauer decomposition. Langlands reduced this to four basic identities, which he called the Main Lemmas. In this paper we formalize the *First Main Lemma*; see [2]¹.

To a quasi-character $\chi_E : E^\times \rightarrow \mathbf{C}^\times$ and a nontrivial additive character $\psi_E : E \rightarrow \mathbf{C}^\times$, Langlands associates a nonzero complex number $\Delta(\chi_E, \psi_E)$, which is essentially a normalized Gauss sum. He also defines the epsilon factor

$$\varepsilon(s, \chi_E, \psi_E) := \Delta\left(|\cdot|_E^{s-\frac{1}{2}} \chi_E, \psi_E\right),$$

where $|x|_E = q_E^{-v_E(x)}$, with q_E the cardinality of the residue field of E . Thus

$$\Delta(\chi_E, \psi_E) = \varepsilon\left(\frac{1}{2}, \chi_E, \psi_E\right).$$

On the other hand, denoting by m and n the conductors of χ_E and ψ_E , respectively, we have

$$\varepsilon(s, \chi_E, \psi_E) = q_E^{-(m+n)(s-\frac{1}{2})} \Delta(\chi_E, \psi_E).$$

Let K/F be a cyclic extension of prime degree ℓ of nonarchimedean local fields. For a quasi-character χ_F and a nontrivial additive character ψ_F , put

$$\chi_K = \chi_F \circ N_{K/F}, \quad \psi_K = \psi_F \circ \mathrm{Tr}_{K/F}.$$

Let

$$S(K/F) = \{\mu \in \mathrm{Hom}_{\mathrm{cont}}(F^\times, \mathbf{C}^\times) : \mu \circ N_{K/F} = 1\}$$

be the group of norm characters.

Langlands's First Main Lemma is the equality

$$(1) \quad \Delta(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F) = \prod_{\mu \in S(K/F)} \Delta(\mu \chi_F, \psi_F).$$

The local proof goes back to Dwork and Langlands. Dwork's 1954 thesis contains the main p -adic calculations [1], while Langlands stated the First Main Lemma and organized the proof by ramification theory and stationary phase [2]. After Deligne found a shorter proof using global functional equations, Langlands stopped preparing a complete version of the local argument.

Dwork used different normalizations and worked in mixed characteristic. Langlands's surviving proof of the First Main Lemma ends before completing the wild endpoint cases, the non-stable wild odd-prime case, and the wild quadratic case; see [Section 2.5](#) for the complete case

¹Chapter 10, entitled "The First Main Lemma", is absent from the surviving manuscript. The identity is nevertheless reproduced in [2, p. 144], where it is attributed to Lemma 10.1.

division. In the companion paper [3], we put these arguments in one notation, complete the remaining calculations, and include the equal-characteristic case. The companion paper gives the mathematical proof; here we describe its formalization in Lean 4 [5], using Mathlib [6].

1.2. The theorem proved in Lean. The proposition expressing the First Main Lemma is `LanglandsFirstMainLemma.FirstMainIdentity`, and the main theorem proving it is `LanglandsFirstMainLemma.firstMainLemma`.

Theorem 1.1 (Formalized First Main Lemma). *Let K/F be a cyclic extension of prime degree between nonarchimedean local fields. For every continuous quasi-character χ_F of $F^\times$ and every nontrivial continuous additive character ψ_F of F , the equality (1) holds.*

The statement has no characteristic-zero hypothesis. Mixed and equal characteristic are treated separately in the final correction calculations. In order to form the product in (1) it is necessary to show that $S(K/F)$ is finite. The formalization proves this via basic ramification theory.

We compared the statement and all the definitions which occur in it with the companion paper. They agree.

1.3. The formal proof. The formal proof follows the companion mathematical proof. We explain below how its choices, conductor conditions, and finite-sum calculations are represented in Lean.

First, both proofs establish nonvanishing of local constants through finite Gauss sums and finite Fourier duality before cancellation. This differs from Langlands's presentation, which starts from the Haar-integral definition and refers to a later part of the argument for nonvanishing.

Second, the mathematical proof repeatedly makes choices of representatives and auxiliary elements. In Lean, the relation between the old and new choices has to be recorded, and the final expression has to be proved independent of them.

Third, several parts of the argument depend on parity, conductor, or ramification conditions. The formalization follows the same case division as the companion proof and records the hypotheses of each case.

Some specific examples of these issues are discussed later, after the corresponding mathematical objects have been introduced.

1.4. Plan of the paper. Section 2 recalls the local constant, the First Main Lemma, and the division of the proof into cases. In Section 3 we give the exact Lean statement and explain its assumptions. Section 4 identifies the local constant, stationary class, and Lamprecht formulas of Section 2.4 with their Lean declarations. The ramification and finite-field results used in the proof are described in Section 5, and the individual cases in Section 6. We record reusable results in Section 7 and describe the released artifact in Section 8.

2. MATHEMATICAL BACKGROUND AND PROOF

We use the notation of the companion paper [3].

2.1. Local fields and conductors. Let E be a nonarchimedean local field. Write

$$\mathcal{O}_E = \{x \in E : v_E(x) \geq 0\}, \quad \mathfrak{p}_E = \{x \in E : v_E(x) \geq 1\},$$

let $k_E = \mathcal{O}_E/\mathfrak{p}_E$ be the residue field, whose cardinality is denoted q_E , and normalize $v_E(E^\times) = \mathbf{Z}$. Put

$$U_E^0 = \mathcal{O}_E^\times, \quad U_E^r = 1 + \mathfrak{p}_E^r \quad (r \geq 1).$$

The quasi-characters are continuous homomorphisms $E^\times \rightarrow \mathbf{C}^\times$, and the additive characters are continuous homomorphisms from the additive group of E to $\mathbf{C}^\times$.

Write $p = \text{char } k_E$. An integer in E means its image under $\mathbf{Z} \rightarrow E$. In particular, $p = 0$ when $\text{char } E = p$. This convention will be important when the same formula is used in both mixed and equal characteristic.

For a quasi-character $\theta : E^\times \rightarrow \mathbf{C}^\times$, the multiplicative conductor $m_E(\theta)$ is the least $m \geq 0$ such that the quasi-character θ is trivial on U_E^m . Thus $m_E(\theta) = 0$ exactly when θ is unramified.

For a nontrivial additive character $\psi : E \rightarrow \mathbf{C}^\times$, the integer $n_E(\psi)$ is defined by requiring $\mathfrak{p}_E^{-n_E(\psi)}$ to be the largest fractional ideal on which ψ is trivial.

These conventions are formalized in Section 4.1 and used throughout the case analysis in Section 6.

2.2. Langlands's local constant. For $z \in \mathbf{C}$, set

$$\text{ph}(z) = \begin{cases} z/|z|, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Let $m = m_E(\theta)$ and $n = n_E(\psi)$, and choose $\gamma \in E^\times$ with

$$(2) \quad \gamma \mathcal{O}_E = \mathfrak{p}_E^{m+n}.$$

Such an element γ is called admissible. For a positive Haar measure du on U_E^0 , Langlands defines

$$(3) \quad \Delta(\theta, \psi) = \theta(\gamma) \text{ph} \left(\int_{U_E^0} \psi(u/\gamma) \theta(u)^{-1} du \right).$$

The value does not depend on the positive normalization of du or on the choice of the admissible element γ .

There is also a finite form of the same constant. Put $Q_m = U_E^0/U_E^m$ and define

$$(4) \quad G(\theta, \psi; \gamma) = \sum_{\bar{u} \in Q_m} \psi(u/\gamma) \theta(u)^{-1}.$$

The summand depends only on the class of u . The quotient Q_m is naturally the unit group of the finite ring $\mathcal{O}_E/\mathfrak{p}_E^m$. Thus (4) is a finite Gauss sum. When $m = 1$ it is the usual Gauss sum over $k_E^\times$.

The finite sum is nonzero, and the local constant has the finite formula

$$(5) \quad \Delta(\theta, \psi) = \theta(\gamma) \text{ph}(G(\theta, \psi; \gamma)).$$

This finite realization and its nonvanishing are [3, Proposition 2.4].

The realizations (3) and (5) are compared formally in Section 4.3 and used in every case of Section 6.

2.3. The First Main Lemma and the first reduction. Recall the setup from Section 1. The First Main Lemma is (1).

It is proved in [3, Corollary 3.11] that

$$(6) \quad |S(K/F)| = |F^\times/N_{K/F}(K^\times)| = [K : F].$$

Local class field theory is not used for this step or elsewhere in the proof.

Define the error quotient

$$(7) \quad \mathcal{E}_{K/F}(\chi_F) = \frac{\Delta(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F)}{\prod_{\mu \in S(K/F)} \Delta(\mu \chi_F, \psi_F)}.$$

The nonvanishing stated in Section 2.2 makes this quotient well defined. The First Main Lemma is equivalent to $\mathcal{E}_{K/F}(\chi_F) = 1$.

If $\mu \in S(K/F)$, then multiplication by μ permutes $S(K/F)$. Therefore

$$(8) \quad \mathcal{E}_{K/F}(\mu \chi_F) = \mathcal{E}_{K/F}(\chi_F).$$

We may consequently replace χ_F by a character of minimal conductor in its $S(K/F)$ -orbit.

The case proofs in Section 6 use this reduction.

2.4. Stationary classes and Lamprecht's formula. Let θ be a quasi-character of $E^\times$, let ψ be a nontrivial additive character of E , and put $m = m_E(\theta)$. Choose an admissible element γ and a positive Haar measure μ_H on U_E^0 . Write

$$(9) \quad I_{\mu_H}(\theta, \psi; \gamma) = \int_{U_E^0} \psi(u/\gamma) \theta(u)^{-1} d\mu_H(u).$$

The decomposition

$$(10) \quad U_E^0 = \coprod_{\bar{a} \in U_E^0/U_E^m} aU_E^m$$

relates this integral to the Gauss sum in (4). The integrand is constant on each coset, since θ is trivial on U_E^m and $\psi((av - a)/\gamma) = 1$ for $v \in U_E^m$. Each coset has measure $c_{\mu_H, m} := \mu_H(U_E^m) > 0$, so

$$(11) \quad I_{\mu_H}(\theta, \psi; \gamma) = c_{\mu_H, m} G(\theta, \psi; \gamma).$$

For $m > 0$, extend $u \mapsto \theta(u)^{-1}$ by zero from the units of $\mathcal{O}_E/\mathfrak{p}_E^m$ to its additive group. Its Fourier coefficients at units are nonzero scalar multiples of $G(\theta, \psi; \gamma)$; those at nonunits vanish by the conductor condition. If G were zero, Fourier inversion would make this nonzero function zero. For $m = 0$, the sum is 1. Thus the integral and the sum are nonzero and have the same phase. Replacing γ by $v\gamma$ with $v \in \mathcal{O}_E^\times$ multiplies G by $\theta(v)^{-1}$, which cancels the change in $\theta(\gamma)$. These are [3, Proposition 2.4 and Lemma 2.6].

For conductor greater than one, Lamprecht's formula reduces the sum further. The stationary class, the formula, and independence of its representative are proved in [3, Lemma 4.2, Theorem 4.5, and Lemma 4.6]. Local constants of conductor zero or one use the formulas in (57). In the wild conductor-one case, the norm characters have higher conductor, so Lamprecht's formula is still needed for their local constants.

Assume $m_E(\theta) > 1$. On a sufficiently deep unit group, the multiplicative character θ can be written in terms of the additive character ψ . More precisely, consider an integer M , an element $\Gamma \in E^\times$ with $\Gamma\mathcal{O}_E = \mathfrak{p}_E^{M+n_E(\psi)}$, and an integer $\lceil m_E(\theta)/2 \rceil \leq r \leq m_E(\theta) - 1$. There is a unique class

$$(12) \quad \text{St}_{\Gamma, r}(\theta) \in \mathfrak{p}_E^{M-m_E(\theta)}/\mathfrak{p}_E^{M-r}$$

with the following property: if $\beta = \beta(\theta)$ represents this class, then

$$(13) \quad \theta(1+x) = \psi(\beta x/\Gamma) \quad (x \in \mathfrak{p}_E^r).$$

The element β is not unique, but its class in (12) is. Indeed, $2r \geq m_E(\theta)$ makes $x \mapsto \theta(1+x)$ an additive character of $\mathfrak{p}_E^r/\mathfrak{p}_E^{m_E(\theta)}$. The perfect pairing defined by ψ identifies that character with the stated quotient class. Every representative satisfies $v_E(\beta) = M - m_E(\theta)$; otherwise the linearization would make θ trivial at a smaller conductor.

Write

$$m_E(\theta) = 2d + \varepsilon > 1, \quad \varepsilon \in \{0, 1\}.$$

Apply this to the special case that $\Gamma = \gamma$ is an admissible element for (θ, ψ) and

$$M = m_E(\theta), \quad r = d + \varepsilon.$$

The stationary class then lies in $\mathcal{O}_E/\mathfrak{p}_E^d$, and every representative β is a unit. Averaging the Gauss integral under $u \mapsto u(1+x)$, for $x \in \mathfrak{p}_E^{d+\varepsilon}/\mathfrak{p}_E^m$, makes the contribution outside $\beta(1 + \mathfrak{p}_E^d)$ vanish by character orthogonality. For even conductor the integrand is constant on this set. For odd conductor its remaining values are indexed by k_E .

If the conductor is even, Lamprecht's formula gives

$$(14) \quad \Delta(\theta, \psi) = \theta(\gamma) \psi(\beta/\gamma) \theta(\beta)^{-1}.$$

If the conductor is odd, the same expression is multiplied by the phase of a finite sum over the residue field. Choose a critical element $\delta \in \mathfrak{p}_E^d \setminus \mathfrak{p}_E^{d+1}$ and define the *Lamprecht critical function* by

$$(15) \quad H_{\theta, \beta, \delta}(\bar{x}) = H_{\theta, \beta}(\bar{x}) = \psi(\delta\beta\bar{x}/\gamma) \theta(1 + \delta\bar{x})^{-1}, \quad \bar{x} \in k_E,$$

where $\tilde{x} \in \mathcal{O}_E$ is any lift of $\bar{x}$. We suppress the critical element δ from $H_{\theta,\beta}$ to lighten the notation; the function itself may depend on δ , but the phase of its sum over k_E does not.

The critical function is well defined and satisfies

$$(16) \quad H_{\theta,\beta}(x+y) = H_{\theta,\beta}(x)H_{\theta,\beta}(y)\psi_0(xy), \quad \psi_0(\bar{z}) = \psi\left(\frac{\beta\delta^2\bar{z}}{\gamma}\right).$$

The character ψ_0 is nontrivial. The identity (16) gives $\left|\sum_{x \in k_E} H_{\theta,\beta}(x)\right|^2 = |k_E|$, so the sum is nonzero. Lamprecht's odd-conductor formula is

$$(17) \quad \Delta(\theta, \psi) = \theta(\gamma)\psi(\beta/\gamma)\theta(\beta)^{-1} \text{ph}\left(\sum_{x \in k_E} H_{\theta,\beta}(x)\right).$$

See [3, Theorem 4.5]. Write

$$\Delta_{2,E}(\theta) = \psi(\beta/\gamma)\theta(\beta)^{-1}, \quad \mathcal{A}_E(\theta; \beta, \gamma) = \theta(\gamma)\Delta_{2,E}(\theta),$$

$$\Delta_{3,E}(\theta) = \begin{cases} 1, & \varepsilon = 0, \\ \text{ph}\left(\sum_{\bar{x} \in k_E} H_{\theta,\beta}(\bar{x})\right), & \varepsilon = 1. \end{cases}$$

Then Lamprecht's formula is

$$(18) \quad \Delta(\theta, \psi) = \theta(\gamma)\Delta_{2,E}(\theta)\Delta_{3,E}(\theta) = \mathcal{A}_E(\theta; \beta, \gamma)\Delta_{3,E}(\theta).$$

The factor $\theta(\gamma)$ records the admissible element, $\Delta_{2,E}(\theta)$ consists of character values at the stationary representative, and $\Delta_{3,E}(\theta)$ is the residue-field factor. For odd conductor, replacing β by $\beta' \equiv \beta \pmod{\mathfrak{p}_E^d}$ changes the last two factors. Put $a = (\beta' - \beta)/(\beta\delta) \in \mathcal{O}_E$. Then

$$H_{\theta,\beta'}(x) = H_{\theta,\beta}(\bar{a})^{-1}H_{\theta,\beta}(x + \bar{a}).$$

Translation preserves the sum, apart from the multiplier $H_{\theta,\beta}(\bar{a})^{-1}$. The stationary factor changes by $H_{\theta,\beta}(\bar{a})$, so their product is unchanged. For even conductor the stationary factor itself is unchanged. This is [3, Lemma 4.6].

To prove the First Main Lemma, one applies (18) to each local constant of conductor greater than one among

$$\Delta(\chi_K, \psi_K), \quad \Delta(\mu, \psi_F), \quad \Delta(\mu\chi_F, \psi_F).$$

The proof then has two parts.

First, one compares the products of the local-field factors $\mathcal{A}_E(\theta; \beta, \gamma)$. If, on the same unit layer,

$$\mu(1+x) = \psi_F(ax/\gamma), \quad \chi_F(1+x) = \psi_F(bx/\gamma),$$

then

$$(\mu\chi_F)(1+x) = \psi_F((a+b)x/\gamma).$$

Thus $a+b$ gives the character identity for $\mu\chi_F$ on the same unit group. If its conductor drops, the stationary class is formed at the new conductor when this is greater than one; otherwise we use the direct formulas for conductor zero or one.

Suppose that χ_F and $\chi_K = \chi_F \circ N_{K/F}$ both have conductor greater than one. Choose admissible elements γ_F, γ_K for (χ_F, ψ_F) and (χ_K, ψ_K) , respectively, and put

$$P_{K/F}(x) = N_{K/F}(1+x) - 1.$$

Choose a stationary depth r_K for χ_K such that $P_{K/F}(x)$ lies in the linearization domain of χ_F for $x \in \mathfrak{p}_K^{r_K}$. For a stationary representative β_F of χ_F on that domain, we have

$$\chi_K(1+x) = \psi_F\left(\frac{\beta_F P_{K/F}(x)}{\gamma_F}\right).$$

The stationary class of χ_K at depth r_K is therefore the unique class whose representatives β_K satisfy

$$(19) \quad \psi_F\left(\frac{\beta_F P_{K/F}(x)}{\gamma_F}\right) = \psi_K\left(\frac{\beta_K x}{\gamma_K}\right), \quad x \in \mathfrak{p}_K^{r_K}.$$

Computing this class in each ramification case gives the compatible representatives needed to compare the factors $\mathcal{A}_F$ and $\mathcal{A}_K$.

Second, one compares the residue-field factors $\Delta_{3,E}(\theta)$. These are finite-field calculations: Hasse–Davenport identities, Hasse–function lifting, quadratic Gauss sums, or the characteristic-two Artin–Schreier identity; the corresponding results are stated in Sections 5.3.1–5.3.6.

The case division in Section 2.5 below records the different ways in which the stationary classes are related under twist and norm, and the finite-field identities needed after that comparison has been made.

2.5. The cases of the proof. Let K/F be a finite Galois extension of nonarchimedean local fields, and write $G = \text{Gal}(K/F)$. For $i \geq 0$, its lower ramification groups are

$$G_i = \{\sigma \in G : \sigma(x) \equiv x \pmod{\mathfrak{p}_K^{i+1}} \text{ for every } x \in \mathcal{O}_K\}.$$

They form a decreasing sequence of subgroups, with $G_i = 1$ for all sufficiently large i .

Now suppose that K/F is cyclic of prime degree ℓ , and put $p = \text{char } k_F$. If K/F is unramified, then $G_i = 1$ for every $i \geq 0$. Otherwise, K/F is totally ramified, so $G_0 = G$. Since a group of prime order has no proper nontrivial subgroup, there is a unique integer $t \geq 0$ such that

$$G_0 = \cdots = G_t = G, \quad G_{t+1} = 1.$$

This integer t is the lower ramification break. In the tamely ramified case, $\ell \neq p$ and $t = 0$; in the wildly ramified case, $\ell = p$ and $t \geq 1$. In the ramified cases, put

$$T = t + 1.$$

By the twist invariance (8), we may choose χ_F of minimal conductor in its $S(K/F)$ -orbit. Write

$$m = m_F(\chi_F) = 2d + \varepsilon, \quad d = \lfloor m/2 \rfloor, \quad \varepsilon \in \{0, 1\}.$$

The proof is divided into the following cases, as in [3, Sections 7–10].

Extension and conductor	Reduction and later references
K/F unramified	For $m = 0$, use the direct formula for an unramified character. For $m = 1$, use the Hasse–Davenport lifting identity. For $m > 1$, compatible stationary representatives reduce the odd-conductor residue-field sum to the Hasse-function lifting identity; when m is even, there is no residue-field sum. See (38), (44), and Sections 5.3.1, 5.3.2, and 6.1.
K/F tamely ramified, ℓ odd	For $m = 0$, use the direct local-field formulas. For $m = 1$, use the Hasse–Davenport product identity. For $m > 1$, every nontrivial $\mu \in S(K/F)$ has conductor one and does not change the stationary class of χ_F ; the remaining odd-conductor residue-field phases satisfy the ramified Hasse-phase identity. See (39), (48), and Sections 5.3.1, 5.3.4, and 6.2.
K/F tamely ramified, $\ell = 2$	For $m = 0$ and $m = 1$, the proof is the same as above. For $m > 1$, the same stationary representative may be used for χ_F , for its twist by the nontrivial member of $S(K/F)$, and for χ_K ; the remaining residue-field identity is evaluated by quadratic Gauss sums. See (47), and Sections 5.3.3 and 6.2.
K/F wild, $m \leq 1$	The conductor-zero case is again direct. In conductor one, Frobenius identifies the Gauss sums for χ_F and χ_K , while (32) determines the remaining factors involving the nontrivial norm characters. The resulting comparison is completed by (66). See Sections 5.2 and 6.3.
K/F wild, $p > 2$, $m \geq 2T$	This is the stable range. Every nontrivial $\mu \in S(K/F)$ has conductor $T \leq d$, so multiplication by μ does not change the stationary class of χ_F . The First Main Lemma reduces to (67). The local-field factors are compared under norm, and the odd-conductor residue-field phases are compared by (48). See Sections 5.3.4 and 6.4.
K/F wild, $p > 2$, $1 < m < 2T$	Here $T > d$, so multiplication by norm characters can change the stationary class. After choosing a generator τ of $S(K/F)$, the stationary classes of τ^j , $\tau^j \chi_F$, and χ_K are computed explicitly. The residue-field phases are evaluated by the quadratic-phase formula and, when $m = T$, by the affine-pencil identity (49); the remaining additive-character value is 1 by [3, Propositions 9.13–9.15]. See Sections 5.2, 5.3.3, 5.3.5, and 6.5.
K/F wild, $p = 2$, $m > 1$	Write $S(K/F) = \{1, \tau\}$. The stationary classes of τ , χ_F , $\tau \chi_F$, and χ_K are compared explicitly. Their residue-field phases are expressed through one characteristic-two quadratic refinement. In the exceptional boundary case, the phases and remaining character values cancel by (56). See Sections 5.3.6 and 6.6.

The ramification, norm, trace, and finite-field statements used in these comparisons are valid in both mixed and equal characteristic. The characteristic distinction appears only in the last correction terms of the wild cases. In equal characteristic those terms vanish in the field. In mixed characteristic their valuations make the additive contribution trivial, except at the wild quadratic odd boundary. There the additive contribution cancels together with the critical-function values and residue-field phases.

The theorem [firstMainLemma_nonarchimedean](#) combines these cases by invoking the corresponding result.

3. THE EXACT LEAN THEOREM

3.1. The formal statement of the First Main Lemma. We now match the terms of the First Main Lemma (1) with their Lean definitions.

The type `ContinuousQuasiCharF` consists of continuous homomorphisms

$$\chi_F : F^\times \longrightarrow \mathbf{C}^\times,$$

and `ContinuousAddCharF` consists of continuous additive characters

$$\psi_F : F \longrightarrow \mathbf{C}^\times.$$

The two pullbacks in the theorem are represented by

$$\text{normQuasiChar } F \ K \ \chi_F = \chi_F \circ N_{K/F}$$

and

$$\text{tracePullbackAddChar } F \ K \ \psi_F = \psi_F \circ \text{Tr}_{K/F}.$$

The type `NormCharacterFK` represents

$$S(K/F) = \{\mu \in \text{Hom}_{\text{cont}}(F^\times, \mathbf{C}^\times) : \mu \circ N_{K/F} = 1\}.$$

The type `LocalConstantFunctionE` consists of functions assigning a complex number to each pair (χ, ψ) of a continuous quasi-character χ of $E^\times$ and a continuous additive character ψ of E . The arguments `DeltaF` and `DeltaK` have types `LocalConstantFunctionF` and `LocalConstantFunctionK`, respectively. In the exported theorem below, `DeltaF` and `DeltaK` are instantiated by the formalized local constants `localConstantF` and `localConstantK`.

The functions `firstMainLeftSide` and `firstMainRightSide` represent the two sides of (1). Their equality is the proposition:

```
def FirstMainIdentity
  (ΔF : LocalConstantFunction F) (ΔK : LocalConstantFunction K)
  (χF : ContinuousQuasiChar F) (ψF : ContinuousAddChar F) : Prop :=
  firstMainLeftSide F K ΔF ΔK χF ψF =
  firstMainRightSide F K ΔF ΔK χF ψF
```

The definition `FirstMainIdentity` is made under the assumption

$$[\text{Finite } (\text{NormCharacter } F \ K)],$$

since its two sides contain products over $S(K/F)$. In the exported theorem, the finiteness of $S(K/F)$ is not assumed separately. The declaration `primeCyclicNormCharacter_finite` proves the finiteness of `NormCharacterFK` for a cyclic extension of prime degree, as required to form the products in the First Main Lemma.

3.2. The exported theorem. The final declaration is `firstMainLemma`:

```
theorem firstMainLemma
  (F K : Type)
  [Field F] [ValuativeRel F] [TopologicalSpace F]
  [IsNonarchimedeanLocalField F]
  [Field K] [ValuativeRel K] [TopologicalSpace K]
  [IsNonarchimedeanLocalField K]
  [Algebra F K] [ValuativeExtension F K]
  [Module.Free F K] [Module.Finite F K]
  (hcyclic : CyclicPrimeExtension F K) :
  let I : PrimeCyclicExtension F K :=
    PrimeCyclicExtension.ofCyclicPrimeExtension F K hcyclic
  let I : Finite (NormCharacter F K) :=
    primeCyclicNormCharacter_finite F K
  ∀ (χF : ContinuousQuasiChar F) (ψF : ContinuousAddChar F),
  ψF ≠ 1 →
    FirstMainIdentity F K (localConstant F) (localConstant K) χF ψF
```

Here `hcyclic` is the hypothesis that K/F is cyclic of prime degree. The first `letI` line makes this same hypothesis available to the ramification lemmas as `PrimeCyclicExtensionFK`. It does not add another assumption.

The second `letI` line uses the finiteness theorem cited in Section 3.1 for `NormCharacterFK`, which represents $S(K/F)$; the finiteness of the latter is proved from `hcyclic` and is not an extra hypothesis.

The theorem then quantifies over a continuous quasi-character χ_F and a continuous additive character ψ_F , with the assumption that ψ_F is nontrivial. Its conclusion is

`FirstMainIdentity F K (localConstant F) (localConstant K) chiF psiF.`

By the definitions in Section 3.1, this is exactly the conclusion (1) in the First Main Lemma.

3.3. The assumptions. The assumptions before `hcyclic` describe the local fields and the extension K/F .

Lean assumption	Mathematical meaning
<code>FieldF</code> , <code>ValuativeRelF</code> , <code>TopologicalSpaceF</code> , <code>IsNonarchimedeanLocalFieldF</code>	F is a nonarchimedean local field with its valuation and topology.
The corresponding assumptions on K	K is a nonarchimedean local field.
<code>AlgebraFK</code>	A specified embedding $F \rightarrow K$.
<code>ValuativeExtensionFK</code>	The valuations on F and K are compatible with this embedding.
<code>Module.FreeFK</code> , <code>Module.FiniteFK</code>	K is finite free as an F -vector space. This is the form used by Mathlib for a finite extension.
<code>CyclicPrimeExtensionFK</code>	K/F is Galois, its Galois group is cyclic, and $[K : F]$ is prime.

The declaration is written with `(FK:Type)`, so it lies in universe zero. This restricts only the universe generality of the Lean statement and has no effect on the local fields considered here.

3.4. The trivial additive character and the phase at zero. Lean defines `localConstantE` for every continuous additive character, so that it is an ordinary total function. Its value at the trivial additive character is set to zero. This convention does not enter the First Main Lemma. The exported theorem assumes that ψ_F is nontrivial, and the proof also establishes the nontriviality of its trace pullback to K .

The phase function is treated similarly:

`phase 0 = 1.`

Before the proof uses a phase arising from a Gauss sum or a Haar integral, that sum or integral is proved to be nonzero. Thus the value assigned at zero is not used in the proof of the theorem.

4. LAMPRECHT'S FORMULA IN LEAN

This section formalizes the local constant and stationary classes recalled in Sections 2.2 and 2.4. These declarations are used in the case proofs of Section 6.

4.1. Exact conductors and admissible elements. Lean represents an exact conductor by a predicate involving a character and an integer. The statement

$$\text{IsMultiplicativeConductor } F \text{ } \chi \text{ } m$$

means that $m = m_F(\chi)$, and

$$\text{IsAdditiveConductor } F \text{ } \psi \text{ } n$$

means that $n = n_F(\psi)$. The structure `LocalQuasiCharDataF` records a quasi-character together with its exact multiplicative conductor, while `LocalAddCharDataF` records a nontrivial additive character together with its exact additive conductor.

The existence and uniqueness theorems for conductors show that every continuous quasi-character χ_F , and every nontrivial continuous additive character ψ_F , determine canonical data of these two types. Accordingly, the proof of `firstMainLemma` replaces `chiF` and `psiF` by

$$\text{canonicalLocalQuasiCharData } F \text{ } \chi F$$

and

$$\text{canonicalLocalAddCharData } F \text{ } \psi F \text{ } h\psi F,$$

where `hpsiF` is the hypothesis $\psi_F \neq 1$.

For

$$\chi : \text{LocalQuasiCharData } F, \quad \psi : \text{LocalAddCharData } F,$$

the type `AdmissibleGammaFchiPsi`, defined by `AdmissibleGamma`, consists of elements $\gamma \in F^\times$ satisfying

$$v_F(\gamma) = m_F(\chi) + n_F(\psi).$$

This is the Lean form of (2), the condition defining admissible elements.

4.2. The finite Gauss sum. For a quasi-character χ of $F^\times$, a nontrivial additive character ψ , and an admissible γ , the theorem `finiteGaussSummandRepresentative_eq_of_congruent` proves that the summand in (4) descends to the unit quotient. The definition `finiteGaussSum` represents the resulting sum:

$$(20) \quad \text{finiteGaussSum } \chi \text{ } \psi \text{ } \gamma = G(\chi, \psi; \gamma).$$

The definition `deltaFinite` then gives

$$(21) \quad \text{deltaFinite } \chi \text{ } \psi \text{ } \gamma = \chi(\gamma) \text{ ph}(\text{finiteGaussSum } \chi \text{ } \psi \text{ } \gamma).$$

The definition `deltaFinite` uses the finite sum; agreement with the integral is proved in Section 4.3.

The theorem `finiteGaussSum_ne_zero` formalizes the Fourier-duality argument in Section 2.4:

$$(22) \quad \text{finiteGaussSum } \chi \text{ } \psi \text{ } \gamma \neq 0.$$

The conductor-zero case is handled separately. For two admissible elements γ, γ' , the theorem `deltaFinite_gamma_independent` proves

$$\text{deltaFinite } \chi \text{ } \psi \text{ } \gamma = \text{deltaFinite } \chi \text{ } \psi \text{ } \gamma'.$$

4.3. Comparison with the Haar integral. Use the integral $I_{\mu_H}(\chi, \psi; \gamma)$ and positive scalar $c_{\mu_H, m}$ from Section 2.4, with $E = F$ and $m = m_F(\chi)$. The definitions `unitHaarIntegrand` and `unitGaussIntegral` represent its integrand and the integral (9). Lean defines

$$(23) \quad \text{deltaIntegral } \mu_H \text{ } \chi \text{ } \psi \text{ } \gamma = \chi(\gamma) \text{ ph}(I_{\mu_H}(\chi, \psi; \gamma)).$$

The declaration `deltaIntegral` keeps the Haar measure and admissible element as arguments.

For the coset decomposition (10), the theorem `integralFinitePositiveScalar_pos` proves positivity of $c_{\mu_H, m}$, and `unitGaussIntegral_eq_positiveScalar_mul_finiteGaussSum` proves (11), with G represented by `finiteGaussSum` in (20). Using (22), the theorems `unitGaussIntegral_ne_zero_of_finiteGaussSum` and `deltaIntegral_eq_deltaFinite` prove nonvanishing of the integral and the equality

$$(24) \quad \text{deltaIntegral } \mu_H \text{ } \chi \text{ } \psi \text{ } \gamma = \text{deltaFinite } \chi \text{ } \psi \text{ } \gamma.$$

Independence of the positive Haar measure and admissible element is proved by `deltaIntegral_measure_independent` and `deltaIntegral_gamma_independent_of_ne_zero`. The exported function `localConstantF` omits these choices. For a nontrivial additive character, the theorems `localConstant_eq_deltaFinite` and `localConstant_eq_deltaIntegral` identify it with (21) and (23). Thus it represents Langlands's local constant (3).

4.4. The stationary class. The declaration `stationaryNumeratorClass` represents the quotient class (12), not a choice of representative. The theorem `stationaryNumeratorClass_existsUnique` proves existence and uniqueness for the character identity (13). For any representative β , the theorem `stationaryNumeratorClass_representative_ord` proves the valuation $v_E(\beta) = M - m_E(\theta)$ established in Section 2.4. In Lamprecht's formula $M = m_E(\theta)$, so the representative is a unit.

4.5. Lamprecht's formula. For even conductor greater than one, the theorem `lamprechtEven` proves (14) for `deltaFinite`. The theorem `deltaIntegral_eq_deltaFinite`, expressed in (24), transfers it to the Haar-integral definition.

For odd conductor greater than one, `lamprechtHasseFunction` represents the critical function (15). The theorem `lamprechtOdd_hasseSum_ne_zero` proves nonvanishing of its sum, and `lamprechtOdd` proves (17), again first for `deltaFinite`.

The theorem `lamprechtOdd_complete_representative_independent` formalizes the cancellation of the two multipliers described in Section 2.4. It states independence of the product in Lamprecht's formula, not independence of its two representative-dependent factors separately.

5. RAMIFICATION AND FINITE-FIELD IDENTITIES

The case proofs use norm and trace calculations in K/F to compare the local constants. For characters of odd conductor greater than one, Lamprecht's formula also leaves a residue-field sum. The finite-field identities below evaluate these sums and the Gauss sums in the conductor-one cases. Our main sources are the norm and trace results of [3, Section 3] and the finite-field identities of [3, Section 5].

5.1. The norm filtration. Let K/F be ramified cyclic of prime degree ℓ , and let t be its lower ramification break, as in Section 2.5. Put

$$T = t + 1, \quad D = (\ell - 1)T,$$

and, for an integer $r \geq 0$, put

$$\Psi(r) = \begin{cases} r, & 0 \leq r \leq t, \\ t + \ell(r - t), & r > t. \end{cases}$$

Thus $\Psi(r)$ is the depth in K corresponding to the depth r in F .

The trace of each fractional ideal is

$$(25) \quad \mathrm{Tr}_{K/F}(\mathfrak{p}_K^s) = \mathfrak{p}_F^{\lfloor (s+D)/\ell \rfloor} \quad (s \in \mathbf{Z}).$$

In particular,

$$(26) \quad n_K(\psi_K) = \ell n_F(\psi_F) + D.$$

For every $r \geq 0$, the norm induces a map

$$N_r : U_K^{\Psi(r)} / U_K^{\Psi(r)+1} \longrightarrow U_F^r / U_F^{r+1}.$$

This map is an isomorphism for $r < t$ and for $r > t$. At $r = t$, its kernel and cokernel both have order ℓ .

The declaration `cyclicPrimeNormFiltration` proves the trace-ideal equality and the norm statements above.

Let $m = m_F(\chi_F)$. For $m > T$, the norm filtration gives

$$(27) \quad m_K(\chi_F \circ N_{K/F}) = \Psi(m - 1) + 1 = \ell m - D.$$

If χ_F has minimal conductor in its $S(K/F)$ -orbit, then

$$(28) \quad m_K(\chi_F \circ N_{K/F}) = m \quad (0 \leq m \leq T).$$

At $m = T$, minimality prevents the conductor from dropping. These statements follow from [3, Proposition 3.9].

For comparison, if K/F is unramified of degree f , then trace and norm preserve the filtration depths:

$$\mathrm{Tr}_{K/F}(\mathfrak{p}_K^s) = \mathfrak{p}_F^s, \quad N_{K/F}(U_K^r) = U_F^r.$$

Hence

$$(29) \quad n_K(\psi_F \circ \mathrm{Tr}_{K/F}) = n_F(\psi_F), \quad m_K(\chi_F \circ N_{K/F}) = m_F(\chi_F).$$

These are formalized by `unramified_additiveConductor_compTrace` and `unramified_multiplicativeConductor_compNorm`. They are used in Section 6.1; the ramified formulas above are used in Sections 5.2, 6.2, 6.3, 6.4, 6.5, and 6.6.

5.2. The norm at the wild break. Let K/F be wildly ramified and cyclic of prime degree p . Then p is the residue characteristic, and the residue fields of F and K are canonically the same finite field; write this field as k . Let $t > 0$ be the unique lower ramification break. Choose a uniformizer ϖ_K of K , put $\varpi_F = N_{K/F}(\varpi_K)$.

Both sides of the critical norm map

$$N_t : U_K^t / U_K^{t+1} \longrightarrow U_F^t / U_F^{t+1}$$

are naturally identified with the additive group of k . If $z \in k$ and $\tilde{z} \in \mathcal{O}_F$ is any lift, the source coordinate is

$$k \longrightarrow U_K^t / U_K^{t+1}, \quad z \longmapsto [1 + \tilde{z} \varpi_K^t],$$

and the target coordinate is defined in the same way using ϖ_F . These coordinates are well defined. In these coordinates the critical norm map is the function $P : k \rightarrow k$ determined by

$$(30) \quad P(z) = \frac{\overline{N_{K/F}(1 + \tilde{z} \varpi_K^t) - 1}}{\varpi_F^t}.$$

To determine the critical norm polynomial P , expand the norm as

$$N_{K/F}(1 + x) = 1 + \mathrm{Tr}_{K/F}(x) + s_2(x) + \cdots + s_{p-1}(x) + N_{K/F}(x),$$

where $s_j(x)$ is the j th elementary symmetric function of the conjugates of x . For $x = \tilde{z} \varpi_K^t$, the ramification estimates put all the terms $s_2(x), \dots, s_{p-1}(x)$ in $\mathfrak{p}_F^{t+1}$. Hence (30) gives

$$(31) \quad P(z) = z^p + cz, \quad c = \frac{\overline{\mathrm{Tr}_{K/F}(\varpi_K^t)}}{\varpi_F^t}.$$

The ramification map determines the kernel of P . Let σ be a generator of $\mathrm{Gal}(K/F) = G_t / G_{t+1}$, and put

$$\lambda = \frac{\overline{\sigma(\varpi_K) - \varpi_K}}{\varpi_K^{t+1}} \in k^\times.$$

The graded norm theorem [3, Theorem 3.1(b)] identifies

$$\ker(U_K^t / U_K^{t+1} \longrightarrow U_F^t / U_F^{t+1})$$

with the image of the ramification map

$$G_t / G_{t+1} \longrightarrow U_K^t / U_K^{t+1}.$$

Under the identification $U_K^t / U_K^{t+1} \simeq k$, this map sends

$$\sigma^j \longmapsto j\lambda, \quad j \in \mathbf{F}_p.$$

Therefore

$$\ker(P) = \lambda \mathbf{F}_p.$$

In particular,

$$(32) \quad P(z) = \prod_{j \in \mathbf{F}_p} (z - j\lambda) = z^p - \lambda^{p-1}z.$$

Comparing (31) and (32) gives the normalized trace identity

$$(33) \quad \frac{\overline{\mathrm{Tr}_{K/F}(\varpi_K^t)}}{\varpi_F^t} = -\lambda^{p-1}.$$

In Lean, the theorem `criticalNormLinearCoefficient_eq_neg_ramificationLambda_pow` proves (33).

The image of the critical norm map also has a simple description. Writing $z = \lambda y$ in (32) gives

$$P(\lambda y) = \lambda^p(y^p - y).$$

The image of $y \mapsto y^p - y$ is the trace-zero subspace of k . Counting dimensions gives

$$(34) \quad \mathrm{im}(P) = \lambda^p \ker(\mathrm{Tr}_{k/\mathbf{F}_p} : k \rightarrow \mathbf{F}_p).$$

The Lean theorem `finiteField_scaledFrobeniusSub_range_eq_scaledTraceKer` proves exactly (34).

This image formula explains why the calculation is needed later. Let $\mu \in S(K/F)$ be nontrivial. The theorem `ramifiedNormCharacter_conductor` gives $m_F(\mu) = t + 1$. Thus the restriction of μ to U_F^t/U_F^{t+1} is a nontrivial additive character of k . Choose the standard additive character

$$(35) \quad \psi_0(z) = \exp\left(\frac{2\pi i}{p} \mathrm{Tr}_{k/\mathbf{F}_p}(z)\right).$$

There is then a unique $a \in k^\times$ such that

$$\mu(1 + \tilde{z} \varpi_F^t) = \psi_0(az) \quad (z \in k).$$

Because μ is trivial on norms, this character is trivial on the subspace in (34). The trace pairing on k is nondegenerate, and the annihilator of $\lambda^p \ker(\mathrm{Tr}_{k/\mathbf{F}_p})$ is the line $\lambda^{-p} \mathbf{F}_p$. Hence

$$a \in \lambda^{-p} \mathbf{F}_p^\times.$$

Thus every nontrivial $\mu \in S(K/F)$ has a unique $j \in \mathbf{F}_p^\times$ such that

$$(36) \quad \mu(1 + \tilde{z} \varpi_F^t) = \psi_0(j \lambda^{-p} z) \quad (z \in k).$$

This describes the restrictions of the nontrivial norm characters to U_F^t/U_F^{t+1} . We use it in the conductor-one calculation of Section 6.3 and the boundary calculation $m = T$ in Section 6.5.

5.3. Finite-field identities. The identities in this subsection are grouped according to the cases listed in Section 2.5. The conductor-zero parts of all cases use only the direct formulas on the local fields and therefore require no finite-field identity from this subsection. Some of these identities are used in more than one of the remaining cases.

5.3.1. Finite Gauss sums: the conductor-one cases. The finite-field formulas below are used in the conductor-one parts of the unramified and tamely ramified cases (Sections 6.1 and 6.2). The same finite Gauss sum also occurs in the wildly ramified conductor-one case in Section 6.3, but the comparison between the local constants over K and F is made by Frobenius rather than by a Hasse–Davenport formula.

Let k be a finite field. Lean uses `FiniteMulChar` for a multiplicative character of $k^\times$, extended by zero at 0, and `FiniteAddChar` for an additive character $k \rightarrow \mathbf{C}^\times$. For a multiplicative character χ and a nontrivial additive character ψ , Langlands’s normalization used throughout the proof is

$$(37) \quad \tau_k(\chi, \psi) = - \sum_{x \in k^\times} \chi(x)^{-1} \psi(x).$$

The Lean definition `langlandsGaussSum` is exactly the sum in (37). The inverse character and the initial minus sign are part of this normalization.

Let k'/k be a finite extension of degree f . The Hasse–Davenport lifting identity `hasseDavenportLift` is

$$(38) \quad \tau_{k'}(\chi \circ N_{k'/k}, \psi \circ \text{Tr}_{k'/k}) = \tau_k(\chi, \psi)^f.$$

This is used when a local Gauss sum over F is compared with the corresponding sum over an unramified extension.

The Hasse–Davenport product identity `hasseDavenportProduct` concerns a product of twists. Suppose that $\ell \mid |k^\times|$ and that η is a multiplicative character of order ℓ . Then

$$(39) \quad \chi(\ell^\ell) \tau_k(\chi^\ell, \psi) \prod_{j=1}^{\ell-1} \tau_k(\eta^j, \psi) = \tau_k(\chi, \psi) \prod_{j=1}^{\ell-1} \tau_k(\chi \eta^j, \psi),$$

where ℓ is viewed as an element of k . This identity is used after a conductor-one local constant has been reduced to a Gauss sum over the residue field.

5.3.2. Hasse functions: the odd higher-conductor cases. This subsection concerns the cases when the quasi-character in a local constant has odd conductor greater than one. In that case Lamprecht’s formula (17) contains the phase of a finite sum over the residue field. The translation and nonvanishing results below are used whenever a change of stationary representative changes the finite function: in Sections 6.2, 6.4, 6.5, and 6.6. The lifting identity (44) is used in the unramified case (Section 6.1).

Let k be a finite field and let $\psi_0 : k \rightarrow \mathbf{C}^\times$ be an additive character. A Hasse function for ψ_0 is a function $H : k \rightarrow \mathbf{C}^\times$ satisfying

$$(40) \quad H(x+y) = H(x)H(y)\psi_0(xy) \quad (x, y \in k).$$

The additive character ψ_0 is therefore the polar character of H . Lean represents this structure by `HasseFunction`.

Write

$$S(H) = \sum_{x \in k} H(x).$$

For $a \in k$, define another Hasse function for the same polar character by

$$H_a(x) := H(x)\psi_0(ax).$$

Summing over $x \in k$ and reindexing by $x \mapsto x+a$ therefore gives

$$(41) \quad S(H_a) = H(a)^{-1} S(H).$$

If ψ_0 is nontrivial, then $S(H) \neq 0$. Taking phases in (41) gives

$$(42) \quad \text{ph}(S(H_a)) = H(a)^{-1} \text{ph}(S(H)).$$

The nonvanishing and the phase identity are proved by `HasseFunction.sum_ne_zero` and `HasseFunction.sumPhase_translate`. Equation (42) is used when changing a stationary representative translates the odd-conductor sum.

Assume that the polar character ψ_0 is nontrivial. There is also a lifting identity for Hasse functions. Let k'/k be a finite extension of degree

$$f = [k' : k].$$

For $x \in k'$, put

$$\omega_{k'/k}(x) = \sum_{\{\sigma, \tau\}} \sigma(x)\tau(x) \in k,$$

where the sum is over unordered pairs of distinct k -embeddings of k' into an algebraic closure. Define

$$(43) \quad H_{k'/k}(x) := H(\text{Tr}_{k'/k}(x)) \psi_0(-\omega_{k'/k}(x)).$$

The Lean definition of this lift is `hasseFunctionLiftValue`. Then $H_{k'/k}$ is a Hasse function on k' for the additive character $\psi_0 \circ \text{Tr}_{k'/k}$, and its sum satisfies

$$(44) \quad - \sum_{x \in k'} H_{k'/k}(x) = \left(- \sum_{x \in k} H(x) \right)^f.$$

The lifting formula (44) is proved by `hasseFunctionLift`. In the unramified case this identity compares the odd-conductor residue-field sum over k_K with the corresponding sum over k_F .

5.3.3. Quadratic phases: the tame quadratic and wild odd-prime cases. This subsection is used in the tamely ramified quadratic case and in both stable and non-stable wild odd-prime ranges: Sections 6.2, 6.4, and 6.5. It evaluates the residue-field sum occurring in Lamprecht's formula when the residue field has odd characteristic.

Let k have odd characteristic, and fix a nontrivial additive character

$$\psi : k \longrightarrow \mathbf{C}^\times.$$

In the applications, the polar character of the Hasse function has the form

$$\psi_a(z) = \psi(az) \quad \text{for some } a \in k^\times.$$

Consider the Hasse function $H : k \rightarrow \mathbf{C}^\times$ for ψ_a :

$$\frac{H(x+y)}{H(x)H(y)} = \psi(axy) \quad (x, y \in k).$$

There is a unique $b \in k$ such that

$$(45) \quad H(x) = \psi\left(\frac{a}{2}x^2 + bx\right) \quad (x \in k).$$

The normal form (45) is proved by `CriticalPolarFunction.eq_quadratic`, and uniqueness of b by `CriticalPolarFunction.affineCoefficient_unique`.

Thus the finite sum in Lamprecht's formula is a quadratic Gauss sum. Define

$$(46) \quad Q_\psi(a, b) = \text{ph} \left(\sum_{x \in k} \psi\left(\frac{a}{2}x^2 + bx\right) \right).$$

This is `quadraticPhase`. Here the sum is nonzero by the Hasse-sum nonvanishing result in Section 5.3.2. Let $\nu : k^\times \rightarrow \{\pm 1\}$ be the quadratic character, defined by

$$\nu(x) = \begin{cases} 1, & x \text{ is a square in } k^\times, \\ -1, & x \text{ is a nonsquare in } k^\times. \end{cases}$$

Completing the square and applying the standard quadratic Gauss-sum identities give

$$(47) \quad Q_\psi(a, b) = \nu(a) \psi\left(-\frac{b^2}{2a}\right) Q_\psi(1, 0), \quad Q_\psi(1, 0)^2 = \nu(-1).$$

The two identities in (47) are proved by `quadraticPhase_eq_basic` and `quadraticPhase_basic_sq`. They evaluate the finite phases remaining in the tame quadratic and wildly ramified odd-prime calculations.

5.3.4. Ramified Hasse phases: the tame odd and stable wild odd-prime cases. This comparison is used in the higher-conductor tame odd-degree case and in the stable wild odd-prime case (Sections 6.2 and 6.4). Recall that Lamprecht's formula has no residue-field phase when the conductor is even.

Let K/F be totally ramified cyclic of odd prime degree ℓ , with lower break t . Write k for the common residue field and suppose

$$m_F(\chi_F) = 2d + 1 > 1.$$

Assume either $t = 0$, or that K/F is wild and $d \geq t + 1$. By (27), the norm pullback $\chi_K = \chi_F \circ N_{K/F}$ also has odd conductor, so Lamprecht's odd-conductor formula applies over both F and K .

Choose an admissible element $\gamma \in F^\times$ for both F and K . The stationary classes of χ_F and χ_K admit the same representative $\beta \in \mathcal{O}_F^\times$. For these choices, the residue-field factors satisfy

$$(48) \quad \Delta_{3,K}(\chi_K) = \Delta_{3,F}(\chi_F)^\ell.$$

The norm calculation relates the two critical functions defining the phases in (48). In odd residue characteristic, one applies the quadratic-phase formulas (45) and (47); in characteristic two, one uses the Hasse-function identity (40). The remaining sign calculation uses $\ell \mid |k^\times|$ in the tame case. The full argument is [3, Theorem 5.6].

For compatible stationary representatives, the phase identity (48) is proved by [ramifiedHasseComparison](#). The norm and finite-field calculations needed for this application are supplied by [ramifiedHassePreparation](#).

5.3.5. The affine-pencil identity. The following finite-field identity is used at the boundary $m = T$ in Section 6.5.

Let k be a finite field of characteristic $p > 2$ and let $\psi : k \rightarrow \mathbf{C}^\times$ be a nontrivial additive character satisfying $\psi(z^p) = \psi(z)$ for all $z \in k$. The character ψ_0 defined in (35) has this property. Suppose $a_0, b_0, \rho, v, \sigma \in k$ satisfy

$$\begin{aligned} \rho &\neq 0, & 1 + j\rho &\neq 0 \quad (j \in \mathbf{F}_p), \\ a_0^p &= 1 - (\rho^{p-1})^{-1}, & b_0^p &= \sigma - \rho^{-1}v. \end{aligned}$$

Then

$$(49) \quad Q_\psi(a_0, b_0) \prod_{j \in \mathbf{F}_p^\times} Q_\psi(j\rho, jv) = \prod_{j \in \mathbf{F}_p} Q_\psi(1 + j\rho, \sigma + jv).$$

The hypotheses ensure that all quadratic coefficients are nonzero. The pairs $(1 + j\rho, \sigma + jv)$ form an affine line in k^2 , which explains the name *affine-pencil identity*. Its Lean theorem is [affinePencil](#).

In the boundary case $m = T$ of Section 6.5, the coefficient pairs that actually occur have an extra common factor $c \in k^\times$: the first pair is (ca_0^p, cb_0^p) , and every other pair is multiplied by c . Proposition 5.8 of [3] shows that this common factor does not change the quotient of quadratic phases. Hence the boundary calculation reduces to the affine-pencil identity (49).

This identity is needed only when T is odd; when T is even, all the corresponding phases are 1.

5.3.6. Characteristic-two quadratic refinements. This subsection supplies the finite-field identities used in the wild quadratic case (Section 6.6).

Let k be a finite field of characteristic two, and put

$$\psi_0(z) = (-1)^{\text{Tr}_{k/\mathbf{F}_2}(z)}.$$

The quadratic expressions of Section 5.3.3 are unavailable in characteristic two. Instead, multiplying a Hasse function with polar character ψ_0 by a suitable additive character gives a quadratic refinement $\mathfrak{q} : k \rightarrow \mathbf{C}^\times$ satisfying

$$(50) \quad \mathfrak{q}(x + y) = \mathfrak{q}(x)\mathfrak{q}(y)\psi_0(xy), \quad \mathfrak{q}(x^2) = \mathfrak{q}(x).$$

If h is another Hasse function with the same polar character, then $h/\mathfrak{q}$ is an additive character. The perfect trace pairing therefore gives a unique $\lambda \in k$ such that $h = h_\lambda$, where

$$(51) \quad h_\lambda(x) = \mathfrak{q}(x)\psi_0(\lambda x), \quad \lambda \in k.$$

Write

$$H(\lambda) = \text{ph} \left(\sum_{x \in k} h_\lambda(x) \right), \quad H_0 = H(0).$$

Translation of the sum gives

$$(52) \quad H(\lambda) = H_0 \overline{\mathfrak{q}(\lambda)}, \quad H_0^2 = \mathfrak{q}(1), \quad \mathfrak{q}(x)^2 = \psi_0(x).$$

In particular,

$$(53) \quad H(\lambda)^2 = h_\lambda(1), \quad H(\lambda+1)H(\lambda) = 1, \quad H(\lambda^2) = H(\lambda).$$

Lean represents the normalized refinement by `CharTwoRefinement`. Its existence is proved by `CharTwoRefinement.exists_charTwoRefinement_translate`, and the unique parametrization by λ is proved by `CharTwoRefinement.existsUnique_affine`.

The boundary calculation uses the following identity for four such functions. Let $\rho \in k^\times$ with $1 + \rho \neq 0$, let $r^2 = \rho$, and suppose $\gamma_0, \gamma, \gamma_1, \gamma' \in k$ satisfy

$$(54) \quad \gamma_1 = \frac{\gamma_0 + \rho\gamma + r}{1 + \rho}, \quad (\gamma')^2 = \frac{\gamma + \rho\gamma_0 + \rho + r}{1 + \rho}.$$

Put

$$(55) \quad \mathcal{R}_{\text{even}} = \frac{H(\gamma')H(\gamma_0)}{H(\gamma)H(\gamma_1)}, \quad B = \frac{\rho^2}{1 + \rho^2}, \quad C = \frac{\rho}{1 + \rho^2}.$$

Then

$$(56) \quad \mathcal{R}_{\text{even}} h_\gamma(B) h_{\gamma_0}(C) \psi_0(C) = \psi_0(c + c^2) = 1, \quad c = (1 + \rho)^{-1}.$$

The first equality follows from the refinement identities (50) and (52); the last uses $\text{Tr}_{k/\mathbb{F}_2}(c + c^2) = 0$. Lean proves this cancellation by `WildQuadraticBoundaryData.value_eq_one`, with the last step supplied by `artinSchreier_phase`.

In Section 6.6, the coefficient relations for the four local critical functions are supplied by [3, Proposition 9.20].

6. THE INDIVIDUAL CASES

For each case of the First Main Lemma (1), we explain the mathematical reduction and then give the Lean declaration proving the corresponding `FirstMainIdentity`. The reductions use the factors in (18) and the ramification and finite-field results of Section 5.

Recall the setup and list of cases of Section 2.5. By the twist invariance (8), we may first replace χ_F by a character of minimal conductor in its $S(K/F)$ -orbit. Write

$$m = m_F(\chi_F) = 2d + \varepsilon, \quad \varepsilon \in \{0, 1\}.$$

In the ramified cases, let t be the lower ramification break and put

$$T = t + 1.$$

Each case below proves

$$\mathcal{E}_{K/F}(\chi_F) = 1,$$

which is equivalent to the First Main Lemma (1).

Let E be a nonarchimedean local field with residue field k_E , $\theta : E^\times \rightarrow \mathbf{C}^\times$ a quasi-character, and $\psi_E : E \rightarrow \mathbf{C}^\times$ a nontrivial additive character. Choose a uniformizer ϖ_E of E .

Formulas for conductors zero and one. If θ has conductor at most one, its residue character is

$$\bar{\theta}(\bar{u}) = \theta(\tilde{u}), \quad \bar{u} \in k_E^\times.$$

Put $n_E = n_E(\psi_E)$. For $v_E(\gamma) = n_E + 1$, define

$$\bar{\psi}_{E,\gamma}(\bar{x}) = \psi_E(\tilde{x}/\gamma).$$

Here tildes denote arbitrary lifts. The character $\bar{\psi}_{E,\gamma}$ is a nontrivial additive character of k_E . When $\gamma = \varpi_E^{n_E+1}$, write $\bar{\psi}_{E,\gamma} = \bar{\psi}_E$. The definition of Δ gives

$$(57) \quad \Delta(\theta, \psi_E) = \begin{cases} \theta(\varpi_E)^{n_E}, & m_E(\theta) = 0, \\ \theta(\gamma) \text{ph}(-\tau_{k_E}(\bar{\theta}, \bar{\psi}_{E,\gamma})), & m_E(\theta) = 1. \end{cases}$$

The minus sign comes from the convention (37). These formulas are formalized by `deltaFinite_conductor_zero` and `deltaFinite_conductor_one`.

Stable twists. Let $\omega : E^\times \rightarrow \mathbf{C}^\times$ be a quasi-character. Write $m_E(\theta) = 2d + \varepsilon > 1$, where $d \geq 0$ is an integer and $\varepsilon \in \{0, 1\}$, and suppose $m_E(\omega) \leq d$. Choose an admissible element γ and a stationary representative β for θ . Then $\omega\theta$ has the same conductor as θ , the same γ is admissible, and the same β is a stationary representative. If the conductor is odd, choose the same critical element δ for both characters. Their critical functions then agree. The stable-twist formula is

$$(58) \quad \Delta(\omega\theta, \psi_E) = \omega(\gamma/\beta)\Delta(\theta, \psi_E).$$

Its Lean counterpart is `stableTwist`.

Products over norm characters. For a finite-order character $\omega : E^\times \rightarrow \mathbf{C}^\times$, [3, Lemma 2.11] gives

$$\Delta(\omega, \psi_E)\Delta(\omega^{-1}, \psi_E) = \omega(-1).$$

Specialize to $E = F$. If ℓ is odd, then every $\mu \in S(K/F)$ has odd order, so $\mu(-1) = 1$. Pairing each nontrivial μ with μ^{-1} , and using $\Delta(1, \psi_F) = 1$, gives

$$(59) \quad \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F) = 1 \quad (\ell \text{ odd}).$$

The inverse identity and the product formula are formalized by `delta_inverse_pair` and `delta_odd_prime_character_product`.

Ramified conductor zero. Suppose that K/F is ramified of prime degree ℓ and $m = 0$. Choose a uniformizer ϖ_K of K , put $\varpi_F = N_{K/F}(\varpi_K)$, and write $n = n_F(\psi_F)$. For each $\mu \in S(K/F)$, the finite-sum formula (5) and the fact that χ_F is unramified give

$$\Delta(\mu\chi_F, \psi_F) = \chi_F(\varpi_F)^{m_F(\mu)+n}\Delta(\mu, \psi_F).$$

Each nontrivial norm character has conductor T , so

$$\prod_{\mu \in S(K/F)} \Delta(\mu\chi_F, \psi_F) = \chi_F(\varpi_F)^{n+(\ell-1)(T+n)} \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F).$$

By (26),

$$n + (\ell - 1)(T + n) = \ell n + (\ell - 1)T = n_K(\psi_K).$$

By (28), $m_K(\chi_K) = 0$. So (57) gives

$$\Delta(\chi_K, \psi_K) = \chi_K(\varpi_K)^{n_K(\psi_K)} = \chi_F(N_{K/F}(\varpi_K))^{n_K(\psi_K)} = \chi_F(\varpi_F)^{n_K(\psi_K)}.$$

Hence

$$\Delta(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F) = \prod_{\mu \in S(K/F)} \Delta(\mu\chi_F, \psi_F),$$

which is the First Main Lemma when $m = 0$.

6.1. Unramified extensions. Suppose that K/F is unramified of prime degree ℓ . Choose a uniformizer ϖ of F , which is also a uniformizer of K , and put $n = n_F(\psi_F)$. The norm is surjective on units, so every $\mu \in S(K/F)$ is unramified. The finite-sum formula (5) gives

$$\Delta(\mu\chi_F, \psi_F) = \mu(\varpi)^{m+n}\Delta(\chi_F, \psi_F), \quad \Delta(\mu, \psi_F) = \mu(\varpi)^n.$$

Taking products over $S(K/F)$ therefore reduces the First Main Lemma to

$$(60) \quad \Delta(\chi_K, \psi_K) = (-1)^{m(\ell-1)}\Delta(\chi_F, \psi_F)^\ell.$$

For $m = 0$, both χ_F and χ_K are unramified. By (57), we have

$$\Delta(\chi_F, \psi_F) = \chi_F(\varpi)^n, \quad \Delta(\chi_K, \psi_K) = \chi_K(\varpi)^n.$$

Since $\chi_K = \chi_F \circ N_{K/F}$ and $N_{K/F}(\varpi) = \varpi^\ell$, (60) follows.

For $m = 1$, let k_K/k_F be the residue field extension, and let $\bar{\chi}$ and $\bar{\psi}$ be the residue characters over F . Since K/F is unramified, (29) gives $n_K(\psi_K) = n$, and the residue characters over K are

$$\bar{\chi}_K = \bar{\chi} \circ N_{k_K/k_F}, \quad \bar{\psi}_K = \bar{\psi} \circ \text{Tr}_{k_K/k_F}.$$

Both χ_F and χ_K have conductor one by (29). Using $\chi_K(\varpi) = \chi_F(\varpi)^\ell$, the conductor-one formula (57) gives

$$\begin{aligned}\Delta(\chi_F, \psi_F) &= \chi_F(\varpi)^{n+1} \text{ph}\left(-\tau_{k_F}(\bar{\chi}, \bar{\psi})\right), \\ \Delta(\chi_K, \psi_K) &= \chi_F(\varpi)^{\ell(n+1)} \text{ph}\left(-\tau_{k_K}(\bar{\chi}_K, \bar{\psi}_K)\right).\end{aligned}$$

The Hasse–Davenport lifting formula (38) states that

$$\tau_{k_K}(\bar{\chi}_K, \bar{\psi}_K) = \tau_{k_F}(\bar{\chi}, \bar{\psi})^\ell.$$

Substituting and taking phases yields

$$\Delta(\chi_K, \psi_K) = (-1)^{\ell-1} \Delta(\chi_F, \psi_F)^\ell,$$

which is (60) for $m = 1$.

For $m > 1$, put $\gamma = \varpi^{m+n}$. By (29), $m_K(\chi_K) = m$ and $n_K(\psi_K) = n$, so γ is admissible over both fields. Choose a stationary representative $\beta \in \mathcal{O}_F^\times$ for χ_F relative to γ . For $y \in \mathfrak{p}_K^{\lceil m/2 \rceil}$, we have

$$P_{K/F}(y) \equiv \text{Tr}_{K/F}(y) \pmod{\mathfrak{p}_F^m},$$

and hence

$$\chi_K(1+y) = \psi_F\left(\frac{\beta \text{Tr}_{K/F}(y)}{\gamma}\right) = \psi_K\left(\frac{\beta y}{\gamma}\right).$$

Thus (19) shows that the same β represents the stationary class of χ_K .

Since $\beta, \gamma \in F$, the norm and trace identities give the relation between local-field factors in (18):

$$(61) \quad \mathcal{A}_K(\chi_K; \beta, \gamma) = \mathcal{A}_F(\chi_F; \beta, \gamma)^\ell.$$

If m is even, Lamprecht’s formula has no residue-field phase, so this proves (60).

If $m = 2d + 1$, Lamprecht’s formula gives

$$\Delta(\chi_E, \psi_E) = \mathcal{A}_E(\chi_E; \beta, \gamma) \text{ph}\left(\sum_{x \in k_E} H_{\chi_E, \beta}(x)\right).$$

To compare the two sums, we may take $\delta = \varpi^d$ in both critical functions. Expanding $N_{K/F}(1 + \delta \tilde{x})$ in (15) to second order identifies $H_{\chi_K, \beta}$ with the Hasse lift of $H_{\chi_F, \beta}$ defined in (43). The lifting formula (44) therefore gives

$$\sum_{x \in k_K} H_{\chi_K, \beta}(x) = (-1)^{\ell-1} \left(\sum_{x \in k_F} H_{\chi_F, \beta}(x) \right)^\ell.$$

Taking phases and combining it with (61) proves (60).

The Lean theorem for this case is `firstMain_unramified_dispatchReady`.

6.2. Tamely ramified extensions. Here F and K have the same residue field k , the break is $t = 0$, and each nontrivial member of $S(K/F)$ has conductor one. The conductor-zero case was proved at the beginning of Section 6. Its Lean counterparts are `firstMain_tameOdd_zero` for odd degree and `firstMain_tameQuadratic_conductor_zero` for degree two.

For $m = 1$, minimality in the $S(K/F)$ -orbit implies that every $\mu\chi_F$ and χ_K has conductor one. Choose $\varpi_F = N_{K/F}(\varpi_K)$ and put $\gamma = \varpi_F^{n_F(\psi_F)+1}$; this is admissible for all these characters. Let $\bar{\chi}, \bar{\psi}$ be the residue characters for χ_F, ψ_F using γ , and let η be the residue character of a generator of $S(K/F)$. Norm and trace induce

$$\bar{\chi}_K = \bar{\chi}^\ell, \quad \bar{\psi}_K(x) = \bar{\psi}(\ell x).$$

Rescaling the Gauss sum for χ_K contributes $\bar{\chi}(\ell^\ell)$. The admissible-element factors agree because $\mu(\gamma) = 1$ and $\chi_K(\gamma) = \chi_F(\gamma)^\ell$. Both sides have the sign $(-1)^\ell$ from the conductor-one formula. Thus (57) reduces the First Main Lemma to the Hasse–Davenport product identity (39); see [3, Proposition 7.2].

For odd ℓ and $m > 1$, stable twisting gives

$$\prod_{\mu \in S(K/F)} \Delta(\mu\chi_F, \psi_F) = \Delta(\chi_F, \psi_F)^\ell,$$

since $\prod_{\mu \in S(K/F)} \mu = 1$. By (59), it remains to prove $\Delta(\chi_K, \psi_K) = \Delta(\chi_F, \psi_F)^\ell$. A common admissible element and stationary representative exist by [3, Proposition 6.8(c)]. With these choices,

$$\mathcal{A}_K(\chi_K; \beta, \gamma) = \mathcal{A}_F(\chi_F; \beta, \gamma)^\ell$$

follows from norm and trace on F . For even conductor both residue-field factors are 1; for odd conductor (48) gives the same power relation. This proves the identity in odd degree, as in [3, Proposition 7.3]. The Lean theorem is `firstMain_tameOdd_of_isDeltaFinite`.

For $\ell = 2$ and $m > 1$, write $S(K/F) = \{1, \tau\}$. Let ν be the quadratic character of $k^\times$, which is the residue character of τ . Choose $\varpi_F = N_{K/F}(\varpi_K) = -\varpi_K^2$ and $\gamma = \varpi_F^{m+n_F(\psi_F)}$. By [3, Proposition 6.8(b)], one stationary representative $\beta \in \mathcal{O}_F^\times$ works for χ_F and χ_K . Stable twisting gives

$$\Delta(\tau\chi_F, \psi_F) = \nu(\bar{\beta})\Delta(\chi_F, \psi_F), \quad \mathcal{A}_K(\chi_K; \beta, \gamma) = \mathcal{A}_F(\chi_F; \beta, \gamma)^2.$$

Consequently the required residue-field identity is

$$(62) \quad \Delta_{3,K}(\chi_K)\Delta(\tau, \psi_F) = \nu(\bar{\beta})\Delta_{3,F}(\chi_F)^2.$$

Put $Q_0 = Q_{\bar{\psi}_F}(1, 0)$, so $Q_0^2 = \nu(-1)$. The quadratic Gauss-sum evaluation (47), applied to the critical functions computed in [3, proof of Proposition 7.4], gives

$$(63) \quad \Delta(\tau, \psi_F) = \nu(2)Q_0, \quad \frac{\Delta_{3,K}(\chi_K)}{\Delta_{3,F}(\chi_F)^2} = \nu(-2\bar{\beta})Q_0.$$

These equalities hold for both parities of m ; the second uses the critical function of χ_K , whose conductor is $2m - 1$. Multiplying (63) and using $Q_0^2 = \nu(-1)$ proves (62). The Lean theorem is `firstMain_tameQuadratic_of_isDeltaFinite`.

6.3. Wild conductors zero and one. Let K/F be wildly ramified of prime degree p , including $p = 2$, with lower break t and $T = t + 1$. The conductor-zero case is the ramified calculation at the beginning of Section 6. Its Lean counterpart is `firstMain_wildEndpoint_zero`.

Suppose $m = 1$. Then χ_K also has conductor one by (28). Choose the admissible elements γ_F, γ_K from [3, Section 9.1], so that the induced residue additive characters are both the canonical ψ_0 of (35). The residue multiplicative characters are $\bar{\chi}$ and $x \mapsto \bar{\chi}(x^p)$. Since Frobenius permutes the residue field and preserves ψ_0 , the two Gauss sums agree. Hence

$$(64) \quad \frac{\Delta(\chi_K, \psi_K)}{\Delta(\chi_F, \psi_F)} = \chi_F\left(\frac{N_{K/F}(\gamma_K)}{\gamma_F}\right).$$

Let τ generate $S(K/F)$, choose its admissible element γ_τ and stationary representative β_τ , and put $c_\tau = \gamma_\tau/\beta_\tau$. The characters τ^j , $1 \leq j < p$, have conductor $T \geq 2$. Thus the stable-twist formula applies to τ^j twisted by χ_F , whose conductor is one. Multiplication over j gives

$$(65) \quad \prod_{j=1}^{p-1} \frac{\Delta(\tau^j\chi_F, \psi_F)}{\Delta(\tau^j, \psi_F)} = \chi_F((-1)^p c_\tau^{p-1}).$$

Here the residues of the integer representatives of j have product $(-1)^p$ in the residue field.

The restriction of τ in (36) and the different determine the remaining character values. For the chosen admissible elements, [3, Lemma 9.1 and its application in Section 9.1] give

$$(66) \quad \frac{N_{K/F}(\gamma_K)}{\gamma_F} \equiv (-1)^p c_\tau^{p-1} \pmod{U_F^1}.$$

Since χ_F is trivial on U_F^1 , (66) identifies the right sides of (64) and (65). This proves the First Main Lemma for $m = 1$. Its Lean counterpart is `firstMain_wildEndpoint_one`.

6.4. The stable wild odd-prime range. Assume that K/F is wildly ramified of prime degree $p > 2$ and that $m = 2d + \varepsilon \geq 2T$, so $d \geq T = t + 1$. Every nontrivial norm character has conductor $T \leq d$. The stable-twist formula (58), the identity $\prod_{\mu \in S(K/F)} \mu = 1$, and (59) reduce the First Main Lemma, just as in the tame odd-degree case, to

$$(67) \quad \Delta(\chi_K, \psi_K) = \Delta(\chi_F, \psi_F)^p.$$

Choose an admissible element $\gamma \in F^\times$ for (χ_F, ψ_F) . The conductor formulas (26) and (27) give

$$v_K(\gamma) = p(m + n_F(\psi_F)) = m_K(\chi_K) + n_K(\psi_K).$$

So γ is also admissible for (χ_K, ψ_K) . Since $d \geq T = t + 1$, [3, Proposition 6.8(c)] allows us to choose a common stationary representative $\beta \in \mathcal{O}_F^\times$ for χ_F and χ_K with this γ .

Because $\beta, \gamma \in F$, the norm and trace identities give

$$\begin{aligned} \mathcal{A}_K(\chi_K; \beta, \gamma) &= \chi_F(\gamma^p) \psi_F(p\beta/\gamma) \chi_F(\beta^p)^{-1} \\ &= \mathcal{A}_F(\chi_F; \beta, \gamma)^p. \end{aligned}$$

Since p is odd, the conductor formula (27) also gives $m_K(\chi_K) \equiv m \pmod{2}$. If both conductors are even, the residue-field factors in Lamprecht's formula are both 1. If they are odd, (48) gives $\Delta_{3,K}(\chi_K) = \Delta_{3,F}(\chi_F)^p$. Together with the equality for $\mathcal{A}_K$, Lamprecht's formula (18) therefore proves (67).

The corresponding Lean theorem is `firstMain_wildStable_of_isDeltaFinite`.

6.5. The non-stable wild odd-prime range. Assume that K/F is wildly ramified of prime degree $p > 2$, with common residue field k , and $1 < m = m_F(\chi_F) < 2T$. Let τ generate $S(K/F)$, with the normalization of [3, Lemma 9.3]. Each nontrivial τ^j has conductor $T > \lfloor m/2 \rfloor$, so the stable-twist hypothesis used in Section 6.4 is not available. The proof compares both the stationary classes and the critical functions for the twists and the norm pullback.

Use the stationary representatives of [3, Proposition 9.8]. Lemma 9.7 of that paper permits replacing Teichmüller coefficients by integer representatives without changing the critical functions. The required congruences for $m > T$ and $m \leq T$ are proved by `wildOdd_highRepresentativeDifference_mem_succ` and `wildOdd_lowRepresentativeDifference_mem_succ`, respectively.

Write

$$R_j = \Delta_{3,F}(\tau^j) \quad (j \neq 0), \quad T_j = \Delta_{3,F}(\tau^j \chi_F), \quad U = \Delta_{3,K}(\chi_K).$$

The residue-field quotient in (7) is

$$(68) \quad \mathcal{R}_{\text{odd}} = \frac{U \prod_{j \in \mathbf{F}_p^\times} R_j}{\prod_{j \in \mathbf{F}_p} T_j}.$$

Each odd-conductor factor is a quadratic phase $Q_{\psi_0}(a, b)$ as in (46); each even-conductor factor is 1. The connection with these chosen representatives is formalized in `Cases/WildOdd/PhasePreparation.lean`.

The coefficient calculations in [3, Lemma 9.6 and Proposition 9.8] give two ways to cancel (68):

$$\begin{aligned} m < T & \quad \left| \begin{array}{l} T_j = R_j \ (j \neq 0), \quad T_0 = U, \\ U = \prod_j T_j, \quad \prod_{j \neq 0} R_j = 1. \end{array} \right. \\ m > T & \end{aligned}$$

The second conclusion also uses the quadratic-phase evaluation (47). These identities are formalized in `Cases/WildOdd/ScalarPhaseProducts.lean`.

When $m = T$, minimality implies that all the characters in the comparison have conductor T . For even T , every residue-field factor is 1. For odd T , use a common admissible element and critical element for τ and χ_F , and use the sum of their stationary representatives for each twist. Writing H_θ for the critical function formed with these choices gives

$$H_{\tau^j \chi_F}(x) = H_{\chi_F}(x) H_\tau(x)^j.$$

Thus the quadratic and linear coefficients of the twists depend affinely on j . The norm calculation in [3, Lemma 9.6 and proof of Proposition 9.9] matches U with the first phase in the homogeneous

affine-pencil identity of Section 5.3.5. The Frobenius substitutions used in this match preserve the sums. That identity yields $U \prod_{j \neq 0} R_j = \prod_j T_j$. The Lean theorem is `wildOdd_boundaryPencil`. Consequently, in all conductor cases,

$$(69) \quad \mathcal{R}_{\text{odd}} = 1.$$

Lean combines the cases in `wildOdd_completeResidualPhaseQuotient_eq_one`.

To compare the character values in $\mathcal{A}_F$ and $\mathcal{A}_K$, write $\beta_\theta, \gamma_\theta$ for the stationary representative and admissible element for $\theta \in \{\tau, \chi_F, \chi_K\}$. The choices in [3, Propositions 6.8(d) and 6.10] give $A \in F^\times$, $u \in K^\times$, and $\mathcal{N} = N_{K/F}(u)$ satisfying

$$(70) \quad \frac{\beta_\tau}{\gamma_\tau} = A, \quad \frac{\beta_{\chi_F}}{\gamma_{\chi_F}} = A\mathcal{N}, \quad \frac{\beta_{\chi_K}}{\gamma_{\chi_K}} = A(\mathcal{N} - u),$$

with $v_K(u) = v_F(\mathcal{N}) = T - m$. The first identity in (70) gives $\tau(1+x) = \psi_F(Ax)$ for $x \in \mathfrak{p}_F^{[T/2]}$, and hence

$$(71) \quad \psi_F(Ax) = 1 \quad (x \in \mathfrak{p}_F^T).$$

The admissible-element values cancel. The remaining multiplicative character values are values on the correction units of [3, Lemmas 9.10–9.11 and Proposition 9.12]; their depths permit replacing them by additive-character values. Proposition 9.13 of that paper combines these with the trace term to give

$$(72) \quad \mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\text{odd}} \psi_F(-AX).$$

Here $X \in F$ is the sum of the trace term and the linearized correction-unit terms defined in that proposition. Proposition 9.14 expresses X as a quotient with a factor p in its numerator. Proposition 9.15 proves that its denominator is nonzero, so $X = 0$ in characteristic p , and proves $X \in \mathfrak{p}_F^T$ in mixed characteristic by valuation estimates. In either case (71) gives $\psi_F(-AX) = 1$. Substituting this and (69) in (72) gives $\mathcal{E}_{K/F}(\chi_F) = 1$.

The representatives $\beta_\tau, \beta_{\chi_F}, \beta_{\chi_K}$ are chosen in `WildOdd/UpperResidualNormalization.lean` and `WildOdd/LowSourceNormalization.lean`. The character linearizations are supplied by `Parameters/HighIntermediate.lean` for $m > T$, and by `WildOdd/CorrectionPreparation.lean` for $m \leq T$. The assembly (72) is proved in `Parameters/PhaseReduction.lean`. The norm identity for X in [3, Proposition 9.14] is proved by `wildOdd_correctionFormula`. The case conclusion is [3, Corollary 9.16], with Lean counterpart `firstMain_wildOdd_nonstable`.

6.6. The wildly ramified quadratic extensions. Assume that K/F is wildly ramified quadratic, with $S(K/F) = \{1, \tau\}$ and common residue field k . Put $m = m_F(\chi_F)$ and $T = t + 1$. Section 6.3 treats $m \leq 1$, so suppose $m > 1$ and retain the minimal-conductor choice of χ_F .

Choose the stationary representatives β_θ and admissible elements γ_θ of [3, proof of Proposition 9.24]. They give $A \in F^\times$, $u \in K^\times$, and $\mathcal{N} = N_{K/F}(u)$ with

$$\begin{aligned} \frac{\beta_\tau}{\gamma_\tau} &= A, & \frac{\beta_{\chi_F}}{\gamma_{\chi_F}} &= A\mathcal{N}, \\ \frac{\beta_{\tau\chi_F}}{\gamma_{\tau\chi_F}} &= A(1 + \mathcal{N}), & \frac{\beta_{\chi_K}}{\gamma_{\chi_K}} &= A(\mathcal{N} - u). \end{aligned}$$

Here $v_K(u) = v_F(\mathcal{N}) = T - m$. The denominator $1 + \mathcal{N}$ is nonzero: this follows from valuation when $m \neq T$ and from minimality when $m = T$. For odd conductors, normalize the critical functions as in [3, Proposition 9.20] and write them as $h_{\gamma_0}, h_\gamma, h_{\gamma_1}, h_{\gamma'}$ for $\tau, \chi_F, \tau\chi_F, \chi_K$, respectively. These are the functions of (51). Their residue-field quotient is

$$\mathcal{R}_{\text{even}} = \frac{\Delta_{3,K}(\chi_K) \Delta_{3,F}(\tau)}{\Delta_{3,F}(\chi_F) \Delta_{3,F}(\tau\chi_F)},$$

with the convention $\Delta_{3,E} = 1$ for even conductor. The choices are recorded in `WildQuadraticSelectedStationaryPairs`.

Put $s = \text{Tr}_{K/F}(u)$ and $w = s/(1 + \mathcal{N})$. The character-value assembly in [3, proof of Proposition 9.24] gives

$$(73) \quad \mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\text{even}} \psi_F(-As) \chi_F(1 - w)^{-1} \tau(1 + w)^{-1}.$$

Stationary linearization, with critical-function values retained for odd conductors, splits the last two character values into additive and residue-field factors. The additive arguments combine into $-AX$, where [3, Lemma 9.17] gives

$$(74) \quad X = s - \mathcal{N}w + w = \frac{2s}{1 + \mathcal{N}}.$$

When $m \neq T$ or T is even. For $m < T$ or even m , [3, Propositions 9.21 and 9.23] give

$$\mathcal{R}_{\text{even}} = 1, \quad \chi_F(1 - w)^{-1} = \psi_F(A\mathcal{N}w), \quad \tau(1 + w)^{-1} = \psi_F(-Aw).$$

For $m > T$ and odd m , the τ -evaluation is the same. Using (53), the other factors are

$$\mathcal{R}_{\text{even}} = H(\gamma)^{-2} = h_\gamma(1)^{-1}, \quad \chi_F(1 - w)^{-1} = \psi_F(A\mathcal{N}w)h_\gamma(1).$$

The two occurrences of $h_\gamma(1)$ cancel. Thus in every case under consideration (73) reduces to

$$\mathcal{E}_{K/F}(\chi_F) = \psi_F(-As + A\mathcal{N}w - Aw) = \psi_F(-AX).$$

In characteristic two, (74) gives $X = 0$. In mixed characteristic, [3, Proposition 9.25] applies the trace estimate (25) to prove $X \in \mathfrak{p}_F^T$. For the present A , the linearization of τ gives (71), so in both characteristics $\mathcal{E}_{K/F}(\chi_F) = 1$.

When $m = T$ and T is odd. This case occurs only in mixed characteristic, by [3, Lemma 9.18]. Here u is a unit. Put $\rho = \bar{u} \in k^\times$; minimality gives $\rho \neq 1$. The coefficients $\gamma_0, \gamma, \gamma_1, \gamma'$ satisfy (54), by [3, Proposition 9.20]. With the notation of (55),

$$B = \frac{\rho^2}{1 + \rho^2}, \quad C = \frac{\rho}{1 + \rho^2}.$$

Proposition 9.23 of the companion paper gives the two residual character factors $h_\gamma(B)$ and $h_{\gamma_0}(C)$, and Proposition 9.25 gives $\psi_F(-AX) = \psi_0(C)$. Consequently

$$(75) \quad \mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\text{even}} h_\gamma(B) h_{\gamma_0}(C) \psi_0(C).$$

The coefficient relations identify (75) with the product in (56). It follows that $\mathcal{E}_{K/F}(\chi_F) = 1$, as in [3, Proposition 9.26].

The cancellation of $\mathcal{R}_{\text{even}}$ with the critical-function values when $m \neq T$ or T is even is proved by `WildQuadraticCoefficientData.phaseRatio_mul_corrections_eq_one_of_not_boundaryOdd`. The two cases are formalized by `wildQuadratic_nonboundary` and `wildQuadratic_boundary`, respectively. Together they prove `firstMain_wildQuadratic`.

6.7. Combining the cases. The seven cases listed in Section 2.5 are mutually exhaustive, and Sections 6.1–6.6 prove the First Main Lemma in each case.

The theorem `firstMainLemma_nonarchimedean` determines which case applies and invokes the corresponding result. Finally, `firstMainLemma` applies this result with `localConstantF` and `localConstantK`, and the conductors of the given characters, proving Theorem 1.1.

7. RESULTS OF POSSIBLE INDEPENDENT USE

The following formalized results may be useful elsewhere.

Result	Lean declaration
The finite Gauss sum occurring in Δ is nonzero, and the finite and Haar-integral realizations of Δ agree; see [3, Proposition 2.4].	<code>finiteGaussSum_ne_zero</code> <code>deltaIntegral_eq_deltaFinite</code>
For a ramified cyclic extension of prime degree, norm is an isomorphism on the graded unit groups away from the break, and $ F^\times/N_{K/F}(K^\times) = [K : F]$ by (6). See [3, Theorem 3.1 and Corollary 3.8].	<code>cyclicPrimeNormFiltration</code> <code>normQuotient_card</code>
At a positive wild break, the critical graded norm is $z \mapsto z^p - \lambda^{p-1}z$, as in (32). See [3, Proposition 3.3].	<code>criticalNormPolynomialValue_exact_of_positiveBreak</code>
The stationary class exists uniquely, and the right-hand side of Lamprecht's odd-conductor formula is independent of the chosen stationary representative; see [3, Lemma 4.2 and Lemma 4.6].	<code>stationaryNumeratorClass_existsUnique</code> <code>lamprechtOdd_complete_representative_independent</code>
For a Hasse function with nontrivial polar character and its lift $H_{k'/k}$, $-\sum_{x \in k'} H_{k'/k}(x) = (-\sum_{x \in k} H(x))^{[k':k]}$ by (44). See [3, Theorem 5.4].	<code>hasseFunctionLift</code>
The affine-pencil identity (49) holds under the hypotheses of Section 5.3.5; see [3, Lemma 5.7].	<code>affinePencil</code>

These declarations use the interfaces of the present project. A future contribution to Mathlib would require comparison with the existing Mathlib interfaces and, where appropriate, a more general statement.

8. ARTIFACT AND REPRODUCIBILITY

The formal artifact [4] is available as release **v1.0.0** of the GitHub repository github.com/fukubillueda-web/epsilon_FML, at commit

b91ad93900a5e6d9207a4e6166ac94bb3bbcf3ed.

It uses Lean v4.32.2 and Mathlib revision

905b95818eb32af7874a58b427f50c1711a5e96c.

The source is distributed under the Apache License 2.0. The file `SOURCE_MAP.md` lists the principal Lean file for each labeled numbered statement in the companion paper.

With Git and `elan/Lean` installed, a fresh checkout can be verified by running

```
git clone https://github.com/fukubillueda-web/epsilon_FML.git
cd epsilon_FML
git checkout v1.0.0
lake exe cache get
lake build
lake env leanchecker --fresh LanglandsFirstMainLemma.Main
```

The project introduces no axioms beyond the standard axioms used by Lean and Mathlib.

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